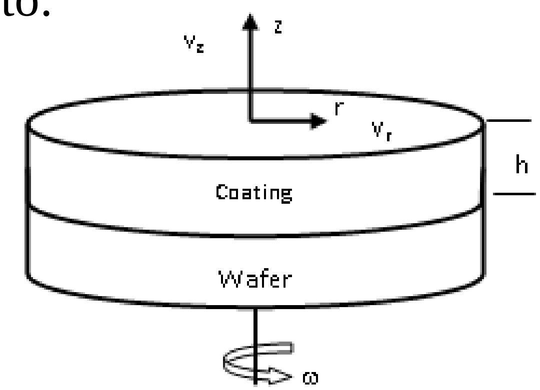
Consider the spin-coating process used to coat silicon wafers with photoresist. The process is designed to produce a very thin, uniform coating by spinning a viscous, Newtonian, liquid onto a substrate (wafer). The process has angular symmetry, the rotation rate is constant, and since the film is thin, there are no real pressure gradients or fluid accelerations and body forces to speak of. The thin film also moves with the substrate as if it were a rigid body, $v_\theta \neq f(z)$

a) Show that the continuity and momentum equations reduce to:

$$-\rho\frac{v_\theta^2}{r}=\mu\frac{d^2 v_r}{dz^2}\,; \qquad \frac{1}{r}\frac{\partial}{\partial r}\left(r v_r\right)+\frac{\partial v_z}{\partial z}=0$$



b) What are the boundary conditions for this problem?
c) Solve the equations for v_r and v_z.
d) The velocity, v_z, at the film/air interface is just the change in film thickness with time. Use this to obtain a differential equation for h, and integrate this equation to obtain the solution as:

$$\frac{1}{2}\left(\frac{1}{h^2}+\frac{1}{h_o^2}\right)=\left(\frac{2\rho\omega^2}{3\mu}\right)t$$

Where ω is the rate of rotation ($v_\theta = 2\pi r\omega$) and h_0 is the initial height of the film.

In this case, the following conditions will hold:

$$v_\theta = Constant,\ g_r = g_\theta = g_z = 0,\ \rho, \mu\ constant$$

$$\frac{\partial}{\partial \theta} = 0, \frac{\partial}{\partial t} = 0, \frac{\partial P}{\partial r} = \frac{\partial P}{\partial \theta} = \frac{\partial P}{\partial z} = 0$$

The continuity and momentum equations in cylindrical coordinates become

$$\frac{\partial \rho}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0 \qquad \frac{1}{r}\frac{\partial}{\partial r}\left(r\, v_r\right) + \frac{\partial v_z}{\partial z} = 0$$

Since the fluid moves essentially as a rigid body, $v_r = f(r, z)$. This film is very thin with respect to the radial or angular dimensions so that

$$v_\theta, v_r \gg v_z$$

Thus since the angular velocity is constant and v_z is small, we only need to deal with the momentum equation for v_r

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} + v_z\frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r}\right) =$$

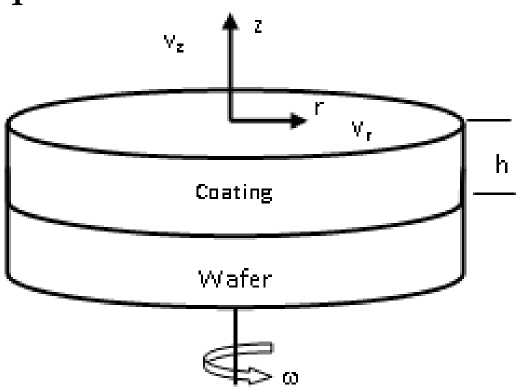
$$-\frac{\partial p}{\partial r} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta}\right] + \rho g_r$$

Neglecting all the obvious terms,

$$\rho\left(v_r\frac{\partial v_r}{\partial r} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r\,v_r)\right) + \frac{\partial^2 v_r}{\partial z^2}\right]$$

$$\mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_r)\right)+\frac{\partial^2 v_r}{\partial z^2}\right]=\rho\left(v_r\frac{\partial v_r}{\partial r}-\frac{v_\theta^2}{r}+v_z\frac{\partial v_r}{\partial z}\right)$$

Since the film is very thin, v_r changes rapidly with the film thickness, much more rapidly than it changes with the radial position. v_z is quite small as compared to the other components



Thus 1st term on LHS is neglected w.r.t. the second term.

For the same reason, 1st and 3rd terms on the rhs can be neglected.

In one case, v_z is very small and in the second $\frac{\partial v_r}{\partial r}$ is also very small.

v_θ is quite large

Thus

$$\mu \frac{d^2 v_r}{d z^2} + \frac{\rho v_\theta^2}{r} = 0$$

The boundary conditions are

$$z = 0;\ v_r = 0$$

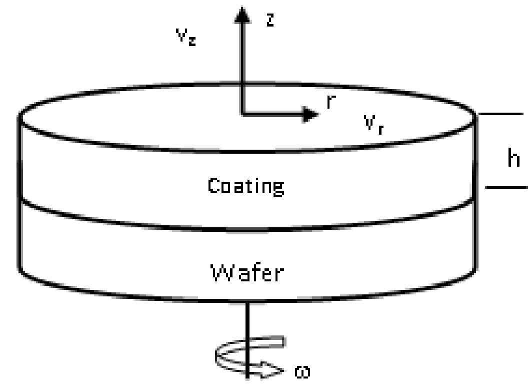
$$z = h;\ \frac{\partial v_r}{\partial z} = 0$$

The coordinate system is such that z = 0 refers to the bottom plate, while z = h is the free surface of the photoresist

Noting that $v_\theta = 2\pi r \omega$, the above equation is solved

$$v_r = \frac{4\pi^2\omega^2}{\gamma} r \left(hz - \frac{z^2}{2} \right)$$

Using the continuity equation



$$\frac{1}{r}\frac{\partial}{\partial r}(r v_r) + \frac{\partial v_z}{\partial z} = 0$$

$$v_z = \frac{4\pi^2\omega^2}{\gamma} \left(\frac{z^3}{3} - h z^2 \right)$$

BC: $v_z = 0$ at $z = 0$

$$\langle v_z \rangle = \frac{1}{h}\int_0^h v_z\, dz = -\frac{\pi^2\omega^2 h^3}{\gamma} = \frac{dh}{dt}$$

$$At \quad t = 0 \quad h = h_0$$

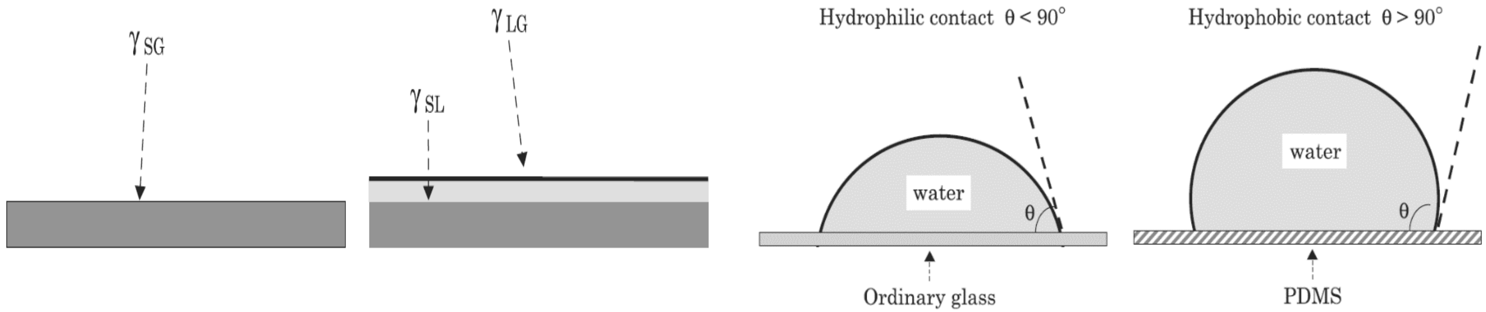
$$\frac{1}{h^2} - \frac{1}{h_0^2} = \frac{2\pi^2\omega^2}{\gamma} t$$

One of the possible answers

This can also be solved by assuming v_z at z = h to be equal to dh/dt

Supplementary Information – in relation to the query about spreading of a droplet on a solid substrate and contact line motion – slip velocity

Wetting: Partial or Total Wetting



A liquid spreads on a substrate in a film **if the energy of the system is lowered by the presence of the liquid film** .

The surface energy per unit surface of the dry solid surface is γ_{SG}; the surface energy of the wetted solid is $\gamma_{SL}+\gamma_{LG}$.

The spreading parameter *S determines the type of spreading (total or partial)*

$$S = \gamma_{SG} - (\gamma_{SL} + \gamma_{LG})$$

If *S > 0, the liquid spreads on the solid surface; if S < 0 the liquid forms a* droplet.